

## Scalability & Future Plan

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| <b>Date</b>          | 2 July 2026                                         |
| <b>Team ID</b>       | EduGenie - Google Gemini Powered Learning Assistant |
| <b>Project Name</b>  | XXXXXX                                              |
| <b>Maximum Marks</b> | 1 Mark                                              |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

#### Current State

|             |                                                            |                                                             |        |  |
|-------------|------------------------------------------------------------|-------------------------------------------------------------|--------|--|
|             | JSON flat-file storage is not safe under concurrent writes | Risk of data loss/corruption under multi-user load          | High   |  |
| <b>S.No</b> | <b>Limitation</b>                                          |                                                             |        |  |
|             | No user authentication or authorization                    | Cannot track individual learning history or restrict access | High   |  |
|             | Single-process Uvicorn server, no caching layer            | Limited throughput under heavy concurrent traffic           | Medium |  |
| 1           | Duplicate router-based and direct API endpoints            | Increases maintenance overhead and API surface confusion    | Medium |  |
| 2           |                                                            |                                                             |        |  |
| 3           |                                                            |                                                             |        |  |

#### Scalability

|             |                    |                                    |                                                                                        |                                                                                     |
|-------------|--------------------|------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|             | User Load          | Single-process Uvicorn, JSON files | Deploy behind Gunicorn + multiple Uvicorn workers; add a reverse proxy / load balancer |                                                                                     |
| <b>S.No</b> | <b>Scalability</b> |                                    |                                                                                        |                                                                                     |
|             | Data Storage       | JSON flat files (data/*.json)      | Migrate to PostgreSQL or MongoDB while preserving the existing db_manager.py interface |                                                                                     |
| 1           | User Load          | Performance                        | No caching; every request hits Gemini/T5                                               | Add response caching for repeated topics; batch quiz generation requests            |
| 2           | Data Storage       | Security                           | No auth, open CORS                                                                     | Add JWT-based authentication, restrict CORS origins, encrypt sensitive data at rest |
| 3           | Performance        |                                    |                                                                                        |                                                                                     |
| 4           | Security           |                                    |                                                                                        |                                                                                     |

### Future Roadmap:

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|-------|-------------------------------|-----------------|-----------------|
|-------|-------------------------------|-----------------|-----------------|

|         |  |  |  |
|---------|--|--|--|
| Phase 2 |  |  |  |
| Phase 3 |  |  |  |
| Phase 4 |  |  |  |

|                                                            |                                                             |        |
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|              |                                          |                                                                                        |
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